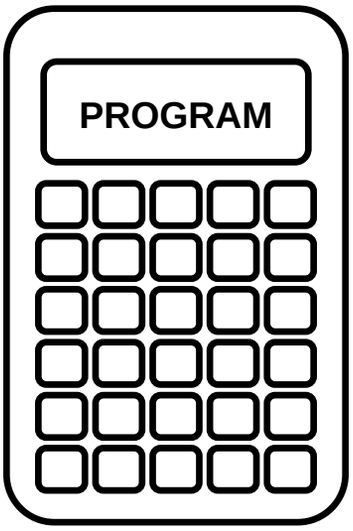
# CASIO fx-50FH II




# S5 PROGRAM LIBRARY

## 中五計算機程式・精選版

7 tested programs · 5 memory-safe packs · 680-byte aware

**Program 01** **一元二次方程：頂點與極值**
Quadratic: Vertex, Optimum and Roots

**Program 02** **聯立二元一次方程**
Simultaneous Linear Equations in Two Unknowns

**Program 03** **一元三次方程**
Cubic Equations

**Program 04** **圓心及半徑**
Centre and Radius of a Circle

**Program 05** **直線與圓的交點**
Intersection Points of a Straight Line and a Circle

**Program 06** **三點定圓**
Centre, Radius and Equation from Three Given Points

**Program 07** **三角形四心**
Centres of a Triangle

**Program 08** **複數聯立方程**
Simultaneous Equations (Complex)

**Program 09** **直線與圓的交點（複數）**
Intersections of a Line and a Circle (Complex)

Built from the canonical tested web catalog · Revised 18 October 2026

# PRGM Command Reference

## 程式模式按鍵速查

| Command | Physical keys | Command | Physical keys |
|---|---|---|---|
| ? | SHIFT 3 1 | ClrMemory | SHIFT 9 1 |
| : / new line | SHIFT 3 3 or EXE | ClrStat | SHIFT 9 1 |
| ◢ output | SHIFT 3 4 | Goto | SHIFT 3 □ □ □ 1 |
| → assign | SHIFT 3 2 | Lbl | SHIFT 3 □ □ □ 2 |
| □ | SHIFT 3 □ 1 | If | SHIFT 3 □ 1 |
| = | SHIFT 3 □ 2 | Then | SHIFT 3 □ 2 |
| ≠ | SHIFT 3 □ 3 | Else | SHIFT 3 □ □ 1 |
| > / < | SHIFT 3 □ □ 1 / 2 | IfEnd | SHIFT 3 □ □ 1 |
| ≥ / ≤ | SHIFT 3 □ □ 3 / 4 | While | SHIFT 3 □ □ □ 1 |
| A B C D | ALPHA + letter key | WhileEnd | SHIFT 3 □ □ □ 2 |
| X Y M | ALPHA + ) , M+ | For / To / Step | SHIFT 3 □ □ □ 1/2/3 |
| Rnd( | SHIFT 0 | Next | SHIFT 3 □ □ □ □ 1 |
| Fix / Sci / Norm | SHIFT MODE □ 1/2/3 | Abs( | SHIFT ) |
| Pol( / Rec( | SHIFT + / SHIFT - | DT | M+ in SD/REG |

Tip: an ordinary minus sign is accepted in Program Studio; use the physical (-) key for negative input on the calculator.

# Program 1

## Quadratic: Vertex, Optimum and Roots
## 一元二次方程：頂點與極值

**用法**

輸入 a、b、c，即時求出頂點、極值及兩個實根。

**INPUT** A=a，B=b，C=c

MODE COMP · 61 bytes

**OUTPUT** 頂點 x、極值、兩個實根

### Program steps

```
?→A
?→B
?→C
B²-4AC→D
-B÷(2A)◢
-D÷(4A)◢
(√(D)-B)÷(2A)◢
-B÷A-Ans
```

### 示範 Example

Solve f(x) = x² + 2x − 15 and describe its graph.

```
TYPE   1  EXE   2  EXE   (-)15  EXE
GET    -1      vertex x-coordinate (h)
       -16     minimum value (k)
       3       first root
       -5      second root

NOTE   The second root appears when the program finishes; press EXE once
       more if the display is still showing the first root. If Math ERROR
       appears at the first root, b² - 4ac < 0 and the roots are not real.
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 2

## Simultaneous Linear Equations in Two Unknowns
## 聯立二元一次方程

**用法**

輸入兩條直線方程的六個係數，即時求出交點。

**INPUT** A,B,C 及 D,X,Y（Ax+By=C、Dx+Xy=Y）

MODE COMP · 51 bytes

**OUTPUT** x、y

### Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
(XC-BY)÷(XA-BD)→M◢
C÷B-AM÷B
```

### 示範 Example

Solve x + y = 1 and 2x - y = 5.

TYPE 1 EXE 1 EXE 1 EXE 2 EXE (-)1 EXE 5 EXE
GET 2 x
-1 y

NOTE The six values are read as A, B, C, D, X, Y. Put the constant of each equation on the right-hand side before you start. If the determinant XA - BD is zero the lines are parallel or identical, and Math ERROR is shown.

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 3

## Cubic Equations
## 一元三次方程

**用法**

毋須先猜因式，直接求一元三次方程的所有實根。

**INPUT** A,B,C,M 為 $Ax^3+Bx^2+Cx+M=0$ 的係數

MODE COMP · 273 bytes

**OUTPUT** 所有不同實根

### Program steps

```
?→A
?→B
?→C
?→M
2×B³÷(27×A³)-B×C÷(3×A²)+M÷A→D
C÷A-B²÷(3×A²)→X
B÷(3×A)→Y
(D÷2)²+(X÷3)³→M
M<0⇒Goto 1
M=0⇒Goto 2
∫((-D÷2+√(M)))+∫((-D÷2-√(M)))-Y◢
Goto 9
Lbl 2
∫((-D÷2))→A
2×A-Y◢
A=0⇒Goto 9
-A-Y◢
Goto 9
Lbl 1
Deg
cos∫¹((3×D÷(2×X))×√((-3÷X)))÷3→M
2×√((-X÷3))→A
A×cos(M)-Y◢
A×cos(M - 120)-Y◢
A×cos(M - 240)-Y◢
Lbl 9
```

### 示範 Example

Factorise $x^3 + 6x^2 + 11x + 6 = 0$.

TYPE 1 EXE 6 EXE 11 EXE 6 EXE
GET -1, -2, -3 the three roots

so $x^3 + 6x^2 + 11x + 6 = (x + 1)(x + 2)(x + 3)$.

NOTE Three pauses means three distinct real roots. One pause means one real root and a complex conjugate pair, which this program does not print.
A repeated root is printed once. Example: 2, 7, 5, 1 gives -0.5, -0.382 and -2.62.

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 4

## Centre and Radius of a Circle
## 圓心及半徑

**用法**

由圓的一般式係數，即時求出圓心及半徑。

**INPUT** A,B,C 為 $x^2+y^2+Cx+By+A=0$ 的係數

MODE COMP · 38 bytes

**OUTPUT** 圓心坐標、半徑

### Program steps

```
?→A
?→B
?→C
-A÷2→X◢
-B÷2→Y◢
√(X²+Y²-C)
```

### 示範 Example

Find the centre and radius of $x^2 + y^2 + 4x - 8y + 3 = 0$.

```
TYPE   4  EXE   (-)8  EXE   3  EXE
GET    -2          x-coordinate of the centre
       4           y-coordinate of the centre
       4.123105626  radius = √17

NOTE   The radius appears when the program finishes. Math ERROR at that step
       means x² + y² + Cx + By + A = 0 is not a real circle.
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 5

## Intersection Points of a Straight Line and a Circle
## 直線與圓的交點

**用法**

把直線代入圓方程，求出兩個交點坐標。

**INPUT** 直線 D,X,Y；圓 C,B,A

MODE COMP · 201 bytes

**OUTPUT** x□, y□, x□, y□

### Program steps

```
?→D
?→X
?→Y
?→C
?→B
?→A
X=0□Goto 1
Y²+CDY+AD²→A
-2XY-CDX+BD²→B
D²+X²→M
(-B-√(B²-4AM))÷(2M)→C
(Y-CX)÷D◢
C◢
(-B+√(B²-4AM))÷(2M)→C
(Y-CX)÷D◢
C◢
Goto 2
Lbl 1
Y÷D→X◢
C²-4(X²+BX+A)→M
(-C+√(M))÷2◢
X◢
(-C-√(M))÷2◢
Lbl 2
```

### 示範 Example

Where does x + 7y = 40 meet x² + y² - 2x - 4y - 20 = 0?

TYPE 1 EXE 7 EXE 40 EXE (-)2 EXE (-)4 EXE (-)20 EXE
GET 5, 5 first intersection point
-2, 6 second intersection point

A vertical line also works: x = 4 and x² + y² - 6x - 6y + 8 = 0
(TYPE 1, 0, 4, -6, -6, 8) gives (4, 6) and (4, 0).

NOTE Input order is the line first, then the circle. A line that misses the circle gives Math ERROR because the discriminant is negative.

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 6

## Centre, Radius and Equation from Three Given Points
## 三點定圓

**用法**

由三個不共線點求圓心、半徑及圓方程。

**INPUT** 六個坐標 x□,y□,x□,y□,x□,y□

**OUTPUT** 圓心、半徑、一般式三個係數

MODE COMP · 166 bytes

### Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
C-A→C
D-B→D
X-A→X
Y-B→Y
(C(X²+Y²)-X(C²+D²))÷2÷(CY-DX)→M
((C²+D²)Y-(X²+Y²)D)÷2÷(CY-DX)→C
A+C→X
B+M→Y
X◢Y◢
√((X-A)²+(Y-B)²)→M◢
-2X◢-2Y◢
X²+Y²-M²
```

### 示範 Example

Find the circle through (5, 2), (2, 3) and (6, -5).

```
TYPE   5  EXE   2  EXE   2  EXE   3  EXE   6  EXE   (-)5  EXE
GET    2, -2        centre (h, k)
       5            radius
       -4, 4, -17   the equation x² + y² - 4x + 4y - 17 = 0

NOTE   Enter the six coordinates in point order. Collinear points give
       Math ERROR because no circle passes through them.
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 7

## Centres of a Triangle
## 三角形四心

**用法**

輸入三個頂點坐標，求形心、內心、垂心及外心。

**INPUT** REG 模式；三個頂點坐標 MODE REG · 385 bytes

**OUTPUT** G、I、H、O 四組坐標

### Program steps

```
ClrStat
?→A
?→B
A,B DT
?→C
?→D
C,D DT
?→X
?→Y
X,Y DT
Σx÷3◢
Σy÷3◢
√((C-X)²+(D-Y)²)+√((X-A)²+(Y-B)²)+√((A-C)²+(B-D)²)→M
(√((C-X)²+(D-Y)²)×A+√((X-A)²+(Y-B)²)×C+√((A-C)²+(B-D)²)×X)÷M◢
(√((C-X)²+(D-Y)²)×B+√((X-A)²+(Y-B)²)×D+√((A-C)²+(B-D)²)×Y)÷M◢
2×(A×(D-Y)+C×(Y-B)+X×(B-D))→M
Pol(((A²+B²)×(D-Y)+(C²+D²)×(Y-B)+(X²+Y²)×(B-D))÷M,((A²+B²)×(X-C)+(C²+D²)×(A-X)+(X²+Y²)×(C-A))÷M)
Rec(X,Y)
Σx-2X◢
Σy-2Y◢
X◢
Y◢
```

### 示範 Example

Take the triangle with vertices O(0, 0), A(24, 0) and B(12, 9).

```
TYPE   0  EXE   0  EXE   24  EXE   0  EXE   12  EXE   9  EXE
GET    12, 3      centroid G
       12, 4      incentre I
       12, 16     orthocentre H
       12, -3.5   circumcentre O

NOTE   This program runs in REG mode and reads the vertices in input order.
       If Math ERROR appears, enter the origin first, then the other two
       vertices: 0, 0, 6, 0, 0, 6 gives G(2, 2), I(1.757, 1.757), H(0, 0), O(3, 3).
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 8

## Simultaneous Equations (Complex)
## 複數聯立方程

**用法**

求係數或答案為複數的二元一次方程組。

**INPUT** CMPLX 模式；A,B,C 及 D,X,Y 可為 a+bi

MODE CMPLX · 51 bytes

**OUTPUT** x、y（複數；按 Re⇄Im 看虛部）

### Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
(XC-BY)÷(XA-BD)→M◢
C÷B-AM÷B
```

### 示範 Example

Solve (1 + i)x + y = 2 and x − y = i.

TYPE 1+i EXE 1 EXE 2 EXE 1 EXE (-)1 EXE i EXE
GET 1 x — a real number, so Re⇄Im shows 0
1 y, real part
-1 y, imaginary part (press SHIFT EXE)

NOTE Run the program in CMPLX mode — i is ALPHA ENG. Every coefficient may be real or complex, and either answer may be complex. At a paused answer, SHIFT EXE turns the display over to the imaginary part.
A zero determinant XA - BD gives Math ERROR.

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

# Program 9

## Intersections of a Line and a Circle (Complex)
## 直線與圓的交點（複數）

**用法**

直線與圓沒有實交點時，求出複數共軛交點。

| | | |
|---|---|---|
| **INPUT** | CMPLX 模式；直線 D,X,Y；圓 C,B,A | MODE CMPLX · 201 bytes |
| **OUTPUT** | x□, y□, x□, y□（可為複數） | |

### Program steps

```
?→D
?→X
?→Y
?→C
?→B
?→A
X=0□Goto 1
Y²+CDY+AD²→A
-2XY-CDX+BD²→B
D²+X²→M
(-B-√(B²-4AM))÷(2M)→C
(Y-CX)÷D◢
C◢
(-B+√(B²-4AM))÷(2M)→C
(Y-CX)÷D◢
C◢
Goto 2
Lbl 1
Y÷D→X◢
C²-4(X²+BX+A)→M
(-C+√(M))÷2◢
X◢
(-C-√(M))÷2◢
Lbl 2
```

### 示範 Example

Where does x + y = 10 meet x² + y² = 25?

```
TYPE   1  EXE   1  EXE   10  EXE   0  EXE   0  EXE   (-)25  EXE
GET    5 + 3.535533906i   x1
       5 - 3.535533906i   y1
       5 - 3.535533906i   x2
       5 + 3.535533906i   y2
```

NOTE  The line misses the circle, so the intersections are a complex conjugate pair. Each answer shows its real part first; SHIFT EXE shows the imaginary part. A line that does meet the circle prints real numbers exactly as the real line-circle program does: 1, 7, 40, -2, -4, -20 gives (5, 5) and (-2, 6).

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.